# Linux Completely Fair Scheduler (CFS) Deep Dive

The Linux Completely Fair Scheduler (CFS), introduced in kernel 2.6.23 by Ingo Molnar, replaces traditional multi-level feedback priority queues with an elegant ideal multi-tasking model. CFS models an 'ideal multi-tasking CPU' where each runnable process receives an equal share of CPU cycles simultaneously. Because hardware consists of discrete processor cores, CFS tracks virtual runtime (vruntime) for each task.

A process's vruntime represents the amount of execution time spent on the CPU scaled inversely by its nice level (priority weight). CFS organizes runnable tasks inside a self-balancing red-black tree (rb-tree) keyed strictly by vruntime. The leftmost node of the tree is always the process with the smallest vruntime--the task most entitled to CPU time next. When a running task consumes CPU cycles, its vruntime increases, and it is re-inserted into the tree. Granularity knobs ensure context switch overhead remains bounded while maintaining sub-millisecond fairness.